

TG500 TPM 2.0 Configuration & Usage Guide

Table of Contents

TG500 TPM 2.0 Configuration & Usage Guide

Applicable models: Tespro TG500 / TG-500 (MT7987A platform)

TPM chip: Infineon SLB9673 (TPM 2.0, I2C interface) Document

version: v1.0 · Tespro Electronics Co., Ltd. (珠海泰易电子技术有限公司)

Contact: Web www.tespro.com · Email Info@tespro.com

1. Overview

The TG500 carries an on-board **Infineon SLB9673** security chip compliant with the **TCG TPM 2.0** specification, connected to the MT7987A host controller over the I2C bus. The core value of a TPM (Trusted Platform Module) is that “**private keys and secrets never leave the chip**”, enabling:

- **Unique device identity authentication:** sign with the in-chip private key to prove “I am this device” to a cloud platform, preventing cloning and spoofing;
- **Key custody:** store SSH / VPN / configuration-encryption keys inside the TPM; even if the flash is read out, the private key cannot be exported;

- **Secure random numbers:** provide hardware-grade true randomness for cryptographic sessions;
- **NV secure storage:** persist small amounts of sensitive data (keys, tokens, serial numbers) in the chip's non-volatile memory;
- **Platform integrity measurement & remote attestation:** record the boot chain in PCRs and prove to a remote party that the firmware has not been tampered with.

2. Hardware Specifications

Item	Parameter
TPM model	Infineon SLB9673 (TPM 2.0)
Bus	I2C (i2c0 controller)
I2C slave address	0x2e (note: NOT the TCG default 0x20)
SCL / SDA	MT7987A GPIO43 / GPIO44 (pinctrl group i2c0_2)
Vendor ID (VID)	0x15D1 (Infineon)
Device ID (DID)	0x001C
Reset method	On-chip RC power-on reset (no dedicated reset GPIO; only a power cycle can reset it)

Device tree node (mt7987a-tespro-TG500.dts):

```
&i2c0 {
    pinctrl-names = "default";
    pinctrl-0 = <&i2c0_2_pins>;
    status = "okay";
```

```
tpm@2e {  
    compatible = "infineon,slb9673";  
    reg = <0x2e>;  
};  
};
```

3. Software Components

The firmware ships with the following components, loaded automatically at boot:

Component	Description
kmod-tpm	Kernel TPM subsystem core
kmod-tpm-tis-i2c	I2C-interface TPM driver (tpm_tis_i2c.ko)
tpm2-tss	TCG software stack (ESAPI/SAPI)
tpm2-tools	User-space command-line toolset (tpm2_*)

Once loaded successfully, two device nodes are created:

Node	Purpose
/dev/tpm0	Raw TPM character device
/dev/tpmrm0	Kernel in-kernel resource manager (recommended; handles object swap-in/out automatically)

Note: this firmware does not enable the user-space resource manager daemon tpm2-abrmd. The kernel's /dev/tpmrm0 already satisfies concurrent access needs, and tpm2-tools uses it by default.

4. Power-on Verification

After the device boots, confirm the TPM is ready as follows:

1) Confirm the device nodes exist

```
ls -l /dev/tpm*
```

Expected: both /dev/tpm0 and /dev/tpmrm0 exist

2) Confirm the driver detects a TPM 2.0

```
dmesg | grep -i tpm
```

Expected: tpm_tis_i2c 0-002e: 2.0 TPM (device-id 0x1C, rev-id 22)

3) Read hardware random bytes (basic functional self-test)

```
tpm2_getrandom --hex 16
```

Expected: 32 hex characters printed, return code rc=0

4) Read chip manufacturer info

```
tpm2_getcap properties-fixed | grep -A1 MANUFACTURER
```

Expected: raw: 0x49465800 ("IFX" = Infineon)

If all four checks pass, the TPM is correctly configured and functional.

5. Common Operations

All commands below have been verified on the TG500.

5.1 Secure random number

```
tpm2_getrandom --hex 32      # get 32 bytes of hardware randomness
```

5.2 Read Platform Configuration Registers (PCR)

```
tpm2_pcrread sha256:0,1,2,3  # read SHA256 values of PCR0~3
```

5.3 Key generation and loading (signing scenario)

```
cd /tmp
```

```
# Create a storage primary key under the owner hierarchy
```

```
tpm2_createprimary -C o -G rsa2048 -c prim.ctx
```

```
# Create a signing child key under the primary key (private key stays in chip)
```

```
tpm2_create -C prim.ctx -G rsa2048:rsassa \  
  -a "sign|sensitivedataorigin|userwithauth" \  
  -u k.pub -r k.priv
```

```
# Load the signing key
```

```
tpm2_load -C prim.ctx -u k.pub -r k.priv -c k.ctx
```

5.4 In-chip signing and verification

```
echo -n "tespro-TG500-identity" > msg.txt
```

```
# Sign with the in-chip private key (-f plain outputs a raw PKCS#1 signature for openssl)
```

```
tpm2_sign -c k.ctx -g sha256 -f plain -o sig.bin msg.txt
```

```
# Export the public key (the private key never leaves the chip)
```

```
tpm2_readpublic -c k.ctx -o pub.pem -f pem
```

```
# Verify with openssl
```

```
openssl dgst -sha256 -verify pub.pem -signature sig.bin msg.txt
```

```
# Expected output: Verified OK
```

5.5 NV secure storage

```
# Define a 16-byte NV space (index 0x1500016)
```

```
tpm2_nvdefine -C o -a "ownerread|ownerwrite" -s 16 0x1500016
```

Write data

```
echo -n "TESPRO-SECRET-01" | tpm2_nvwrite -C o -i - 0x1500016
```

Read data back

```
tpm2_nvread -C o -s 16 0x1500016 | hexdump -C
```

Delete the NV space

```
tpm2_nvundefine -C o 0x1500016
```

5.6 Endorsement Key (EK)

```
tpm2_createek -G rsa2048 -u ek.pub -c ek.ctx # create the endorsement key
```

Remote attestation (quote / AK) is an advanced attestation feature involving authorization-policy configuration on the endorsement hierarchy. To enable it, please contact Tespro technical support.

6. Typical Application: Device Identity Signing

Below is a minimal end-to-end example of “signing a device identity with the TPM private key and verifying it on the server”:

— Device side —

```
tpm2_createprimary -C o -G rsa2048 -c prim.ctx
```

```
tpm2_create -C prim.ctx -G rsa2048:rsassa \
  -a "sign|sensitivedataorigin|userwithauth" -u k.pub -r k.priv
```

```
tpm2_load -C prim.ctx -u k.pub -r k.priv -c k.ctx
```

```
tpm2_readpublic -c k.ctx -o pub.pem -f pem # report public key to server/clo  
ud
```

```
echo -n "$DEVICE_ID:$TIMESTAMP" > challenge.txt # challenge issued by the ser  
ver
```

```
tpm2_sign -c k.ctx -g sha256 -f plain -o sig.bin challenge.txt
```

```
# send sig.bin back to the server
```

```
# — Server side —
```

```
openssl dgst -sha256 -verify pub.pem -signature sig.bin challenge.txt
```

```
# "Verified OK" proves the signature was produced by the TG500 device holding  
the matching private key
```

Because the private key always remains inside the TPM and cannot be exported, even a complete copy of the device flash cannot let an attacker forge a signature, thereby achieving reliable device identity authentication.

7. Precautions & Maintenance

1. **The I2C address is 0x2e:** the SLB9673 responds at 0x2e on this board, not the TCG-spec default 0x20. Always use 0x2e in the device tree.
2. **Cold-boot probe tolerance:** the firmware adds cold-boot retry and deferred-probe (-EPROBE_DEFER) logic to the TPM driver; a normal cold boot completes detection within about 9 seconds.
3. **Reset requires a power cycle:** the SLB9673 uses an on-chip RC power-on reset with no dedicated reset pin. If the TPM enters an abnormal state (command-level unresponsive), rmmod/modprobe or a warm reboot will NOT help — you **must power off for a few seconds and power back on** to recover.
4. **Avoid abnormal power loss:** a sudden power cut while the TPM is performing key operations has a very low probability of leaving the chip in a state that needs a power cycle to recover. Perform bulk key operations under stable power.

5. **Key persistence:** the primary key from tpm2_createprimary changes after a TPM restart (it is non-persistent). To fix a key, use tpm2_evictcontrol to make it a persistent handle.
-

8. Troubleshooting

Symptom	Possible cause	Action
ls /dev/tpm* shows no node	Driver not loaded / wrong DTS address	Check dmesg <code>\ grep tpm</code> ; confirm DTS uses tpm@2e
dmesg shows 1.2 TPM and probe failed -62	TPM is in a command-level abnormal state	Power off for a few seconds, then power back on
tpm2_sign reports 0x19C	Tried to sign with a storage key	The signing key must carry the sign attribute (see 5.3)
tpm2_sign reports 0x1D5	Input is plaintext but -d was added	Drop -d when signing plaintext; use -d only for digests
openssl verify fails with invalid padding	Signature is in TSS-wrapped format	Add -f plain when signing to output a raw signature
tpm2_createak returns rc=5	Endorsement-hierarchy authorization policy restriction	Contact Tespro technical support for configuration

Appendix: Command Quick Reference

```
tpm2_getrandom --hex N      # hardware random number
tpm2_getcap properties-fixed # fixed chip properties
tpm2_pcrread sha256:0,1,2,3 # read PCRs
tpm2_createprimary -C o -G rsa2048 -c prim.ctx # primary key
tpm2_create -C prim.ctx -G rsa2048:rsassa -a "sign|sensitive|data|origin|user|with|auth" -u k.pub -r k.priv
tpm2_load -C prim.ctx -u k.pub -r k.priv -c k.ctx
tpm2_sign -c k.ctx -g sha256 -f plain -o sig.bin msg.txt
tpm2_readpublic -c k.ctx -o pub.pem -f pem
tpm2_nvdefine / nvwrite / nvread / nvundefine # NV storage
tpm2_createek -G rsa2048 -u ek.pub -c ek.ctx # endorsement key
```